Part 2 left you here



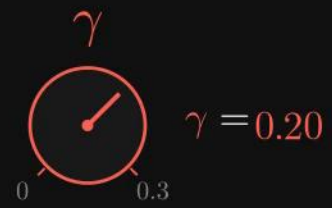
χ_{zz}?



Positive coupling: neighbours prefer to point opposite.

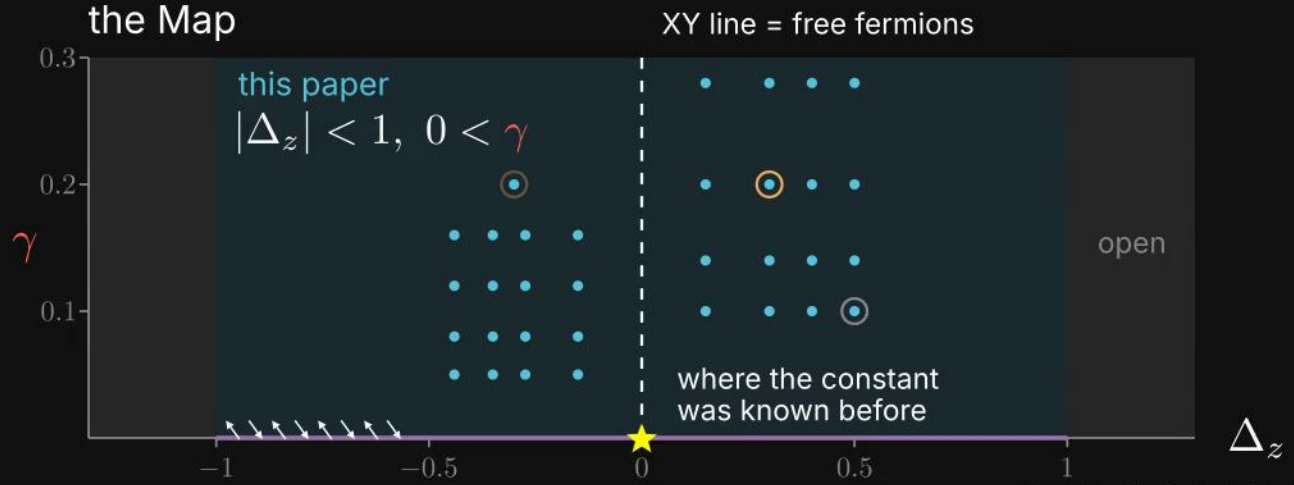
length of a rod = how much the pair cares about that axis

$$H = \sum_j \left[(1+\gamma)\,\sigma_j^x \sigma_{j+1}^x + (1-\gamma)\,\sigma_j^y \sigma_{j+1}^y + \Delta_z\,\sigma_j^z \sigma_{j+1}^z \right]$$



? One knob, gamma. Which of its consequences does the meter care about?

rod lengths 1 + gamma, 1 - gamma, Delta_z: closed form
home coupling Delta_z = 0.3, gamma = 0.2: rods 1.2, 0.8, 0.3

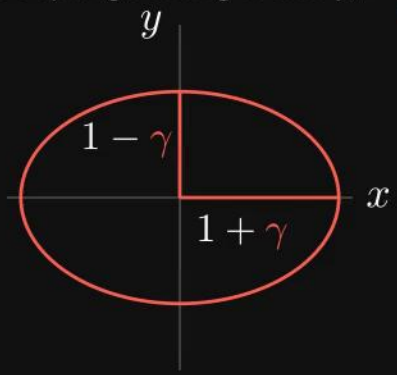


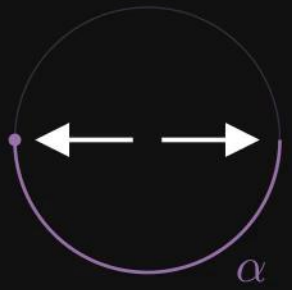
paper: fig1_points.json, new_chi.json, attr_results.json (dots); home Delta_z = 0.3, gamma = 0.2 (val2.json: Delta_z = -0.3, gamma = 0.2; chi_conv.json: Delta_z = 0.5, gamma = 0.1)

BLACK BOX · Jordan-Wigner (XY line = free fermions)

- On the line Δz = 0 a change of variables turns the spins into fermions that do not interact; stated, not shown.
- Two exact results live on that line and feed 3.7 and 4.5: the kink's inverse size and the number under Part 2's question mark.
- Off the line the fermions interact; nothing in these talks relies on the fermion picture.

couplings along x and y, $\gamma = 0.20$





$\langle H \rangle$ per bond, product state

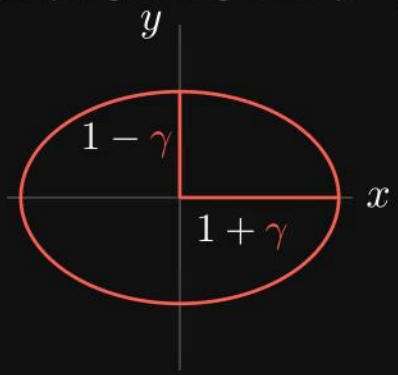
$$-1.20$$

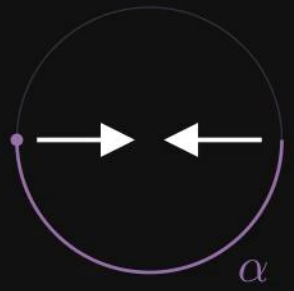
the energy knows where x is

a product state, not the ground state

closed form: -1 - gamma cos 2 alpha (readout); 1 + gamma, 1 - gamma (axes)
talknum: product-state energy, L = 8 ring, Delta_z = 0.3, gamma = knob; not the ground state

couplings along x and y, $\gamma = 0.20$





$\langle H \rangle$ per bond, product state

-1.20

the energy knows where x is

a product state, not the ground state

$$\sigma^x\sigma^x - \sigma^y\sigma^y = 2(\sigma^+\sigma^+ + \sigma^-\sigma^-)$$

$$[H, Q] \neq 0 \iff \gamma \neq 0$$

the red-corner factory



$Q = 0$

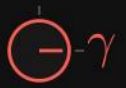
Q = Part 2's Q_A with the window removed

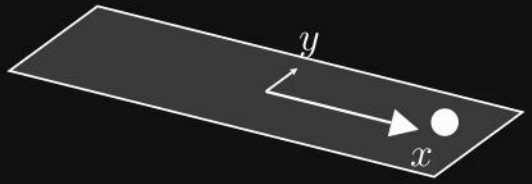
γ moves the chain's charge in steps of two.

closed form: -1 - gamma cos 2 alpha (readout); 1 + gamma, 1 - gamma (axes)
talknum: product-state energy, L = 8 ring, Delta_z = 0.3, gamma = knob; not the ground state

explicit

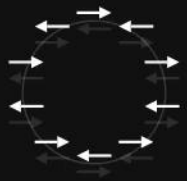
the tilted table





γ tilts the table: the rule itself prefers x

$$[H, Q] \neq 0$$



a finite ring returns the cat:
both falls at once;
we pin one (Ledger)

spontaneous

the falling pencil



the rule is even-handed between $+x$ and $-x$



same energy same energy

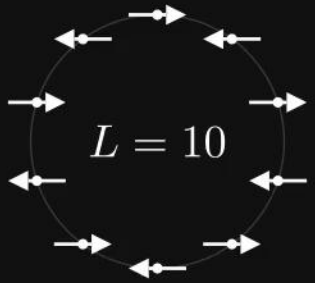
$$F = \prod_j \sigma_j^z$$

$$[H, F] = 0, \;\; F|\mathrm{GS}\rangle \neq |\mathrm{GS}\rangle$$

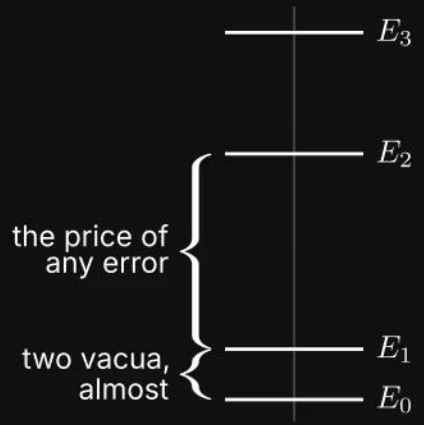
Tilted table (explicit): γ tilts the rule itself; $[H, Q] \neq 0$.
Falling pencil (spontaneous): the rule is even-handed,
the state picks a side; $[H, F] = 0, \;\; F|\mathrm{GS}\rangle \neq |\mathrm{GS}\rangle$.
The $\frac{1}{2} \log \ell$ growth needs the tilted table; the numerics sit in one fall of the pencil.

$E_2 - E_0 > 0, \quad E_1 - E_0 \to 0$ (two vacua)

a ring of ten spins



the energy ladder



$E_2 - E_0 = 1.748$

ring toy, pinned, L = 10; Delta_z = 0.3, gamma = 0.2

$E_1 - E_0 = 0.359$

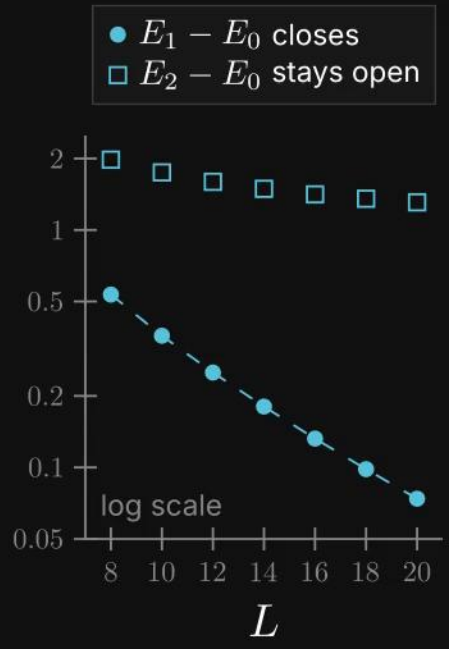
γ



$\gamma = 0.20$

$\Delta_z = 0.3$

energies in units of the couplings; never compared with M



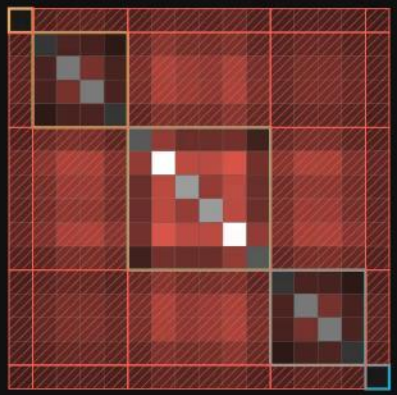
ring toy, pinned, L = 8 to 20; Delta_z = 0.3, gamma = 0.2 (ladder against gamma at L = 10; energies in units of the couplings)

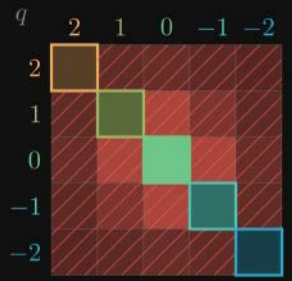


$$\langle \sigma_j^x \rangle = (-1)^j m_x, \quad \langle \sigma_j^z \rangle = 0 \quad m_x = 0.767$$

the order parameter

along z every spin is a fair coin; the stripes live in the red corners





block weights
(sum of squared cells)

$\lvert\Delta q\rvert = 0$	0.383
$\lvert\Delta q\rvert = 1$	0.347
$\lvert\Delta q\rvert = 2$	0.050
$\lvert\Delta q\rvert = 3$	0.003

all blocks shaded; $\lvert\Delta q\rvert = 1$ dominates

two apart (even): $F = \prod_j \sigma_j^z$

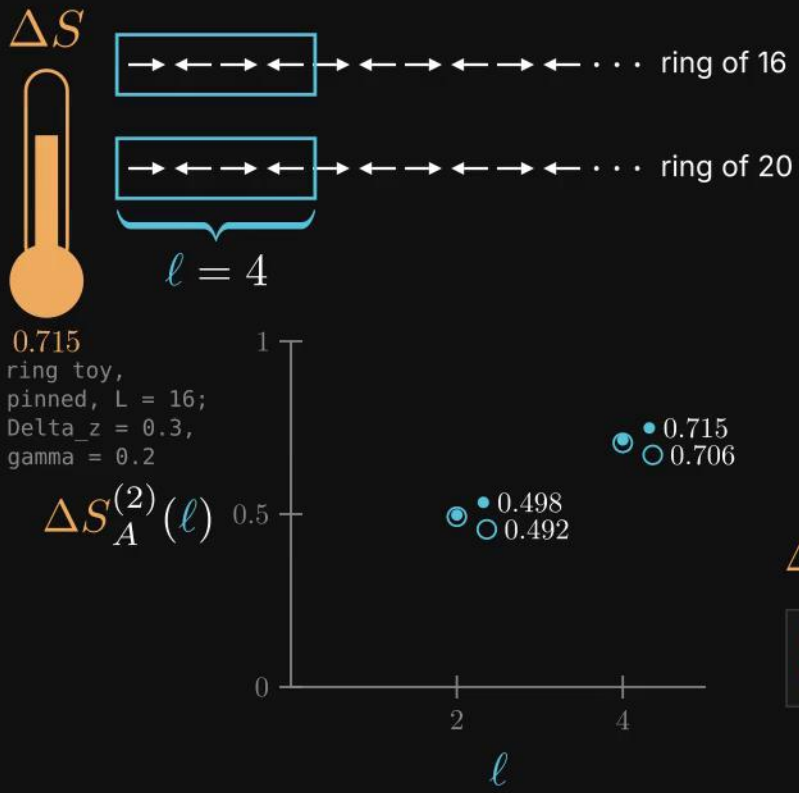
allowed by the symmetry γ leaves behind: the charge moves in steps of two, so even the symmetric, unfallen ring state has them already

one apart (odd):

a needle along x is a corner between up and down, one spin at a time: these need the fall of the pencil

Ledger: pinned = the combination of the two lowest ring states with the largest overlap with the Neel-x product state

ring toy, pinned, L = 16; Delta_z = 0.3, gamma = 0.2
(l = 4 window; L = 10 and L = 20 values in the notes)



? PAUSE AND PONDER

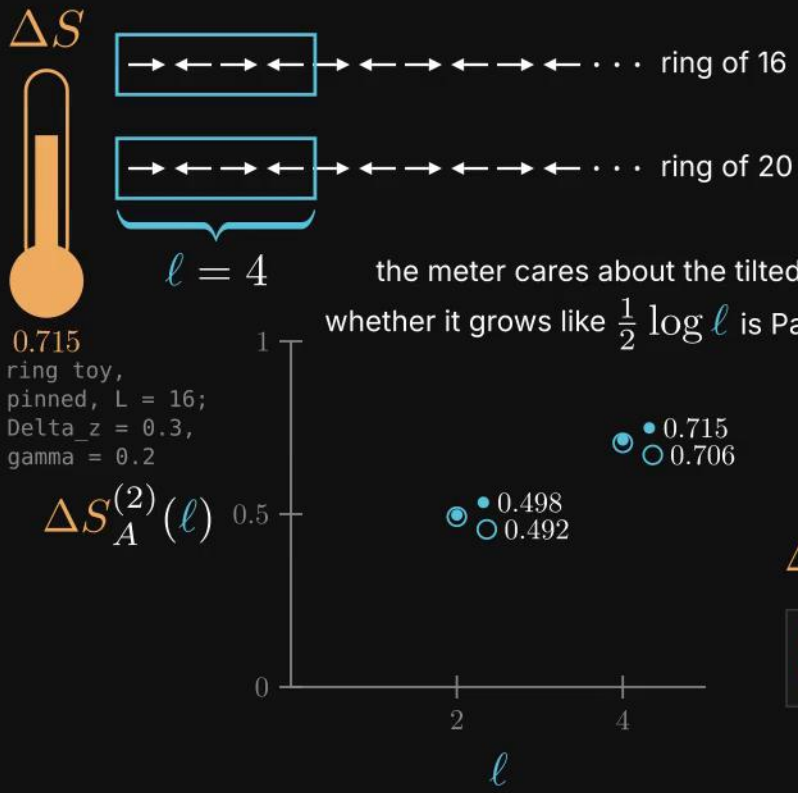
Turn the knob off (γ = 0, same Δz). What does the meter read on the symmetric ground state?

Vote: Still about one half | Exactly zero, for every window | It depends on the ring size

$\Delta S_A^{(2)}(\ell)$ from ED ρ_A

- ring of 16
- ring of 20

ring toy, pinned, L = 16, 20; Delta_z = 0.3, gamma = 0.2
(even l <= L/4; the gamma = 0 column is exact, not a sweep)



the meter cares about the tilted table;
whether it grows like $\frac{1}{2}\log \ell$ is Part 4's test

ring toy, pinned, L = 16, 20; Delta_z = 0.3, gamma = 0.2
(even l <= L/4; the gamma = 0 column is exact, not a sweep)

? PAUSE AND PONDER

Turn the knob off (γ = 0, same Δz). What does the meter read on the symmetric ground state?

Vote: Still about one half |
Exactly zero, for every window |
It depends on the ring size

symmetric ground state
$\gamma = 0$, same Δ_z

ℓ	ΔS
$\ell = 2$	0
$\ell = 4$	0
every ℓ	0

flat table: one definite charge, no corners between sectors

closed form; checked on the ring toy, unpinned, L = 16; Delta_z = 0.3, gamma = 0; l = 2, 4

does spin r remember spin 0's z?

$\gamma = 0.05$

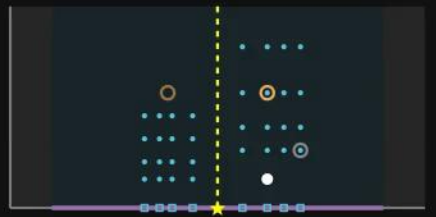
$M = 0.019552$

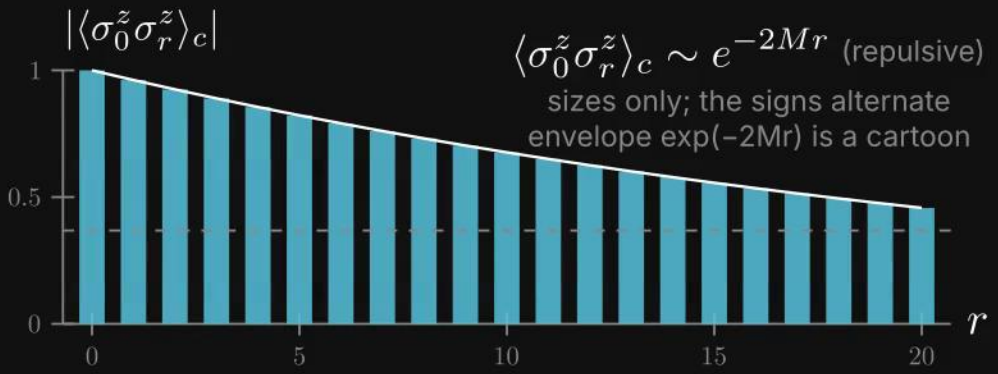
paper: jkm_exact.py M_exact
Delta_z = 0.3; gamma = 0.2, 0.1, 0.05





the Map





at $\gamma = 0$ the memory is infinite:
the gapless edge of the Map

pair-correlation length $1/(2M)$

$= 25.6$ sites

kink size $1/M$

M is an inverse length:
$1/M$ counts sites

does spin r remember spin 0's z?

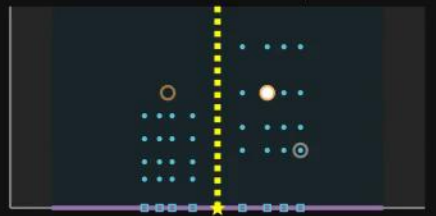


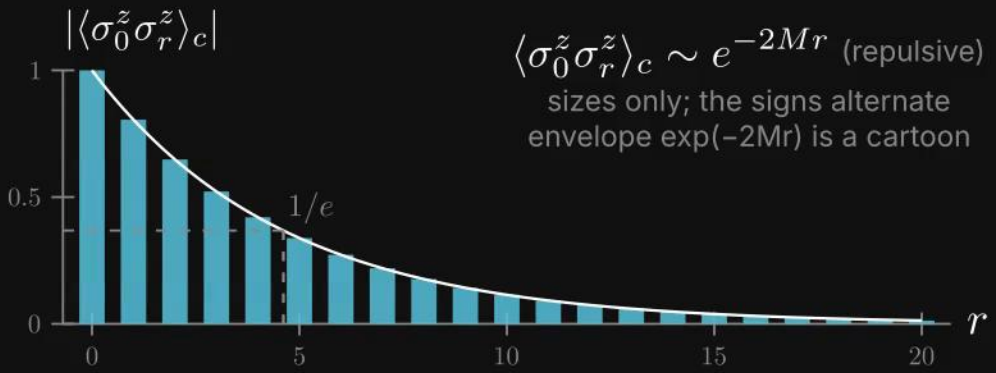


$M = 0.10836$

paper: jkm_exact.py M_exact
Delta_z = 0.3; gamma = 0.2, 0.1, 0.05

XY line $M = \operatorname{atanh}\gamma$





$\langle\sigma_0^z\sigma_r^z\rangle_c \sim e^{-2Mr}$ (repulsive)

sizes only; the signs alternate
envelope exp(−2Mr) is a cartoon

pair-correlation length $1/(2M)$
$= 4.6$ sites

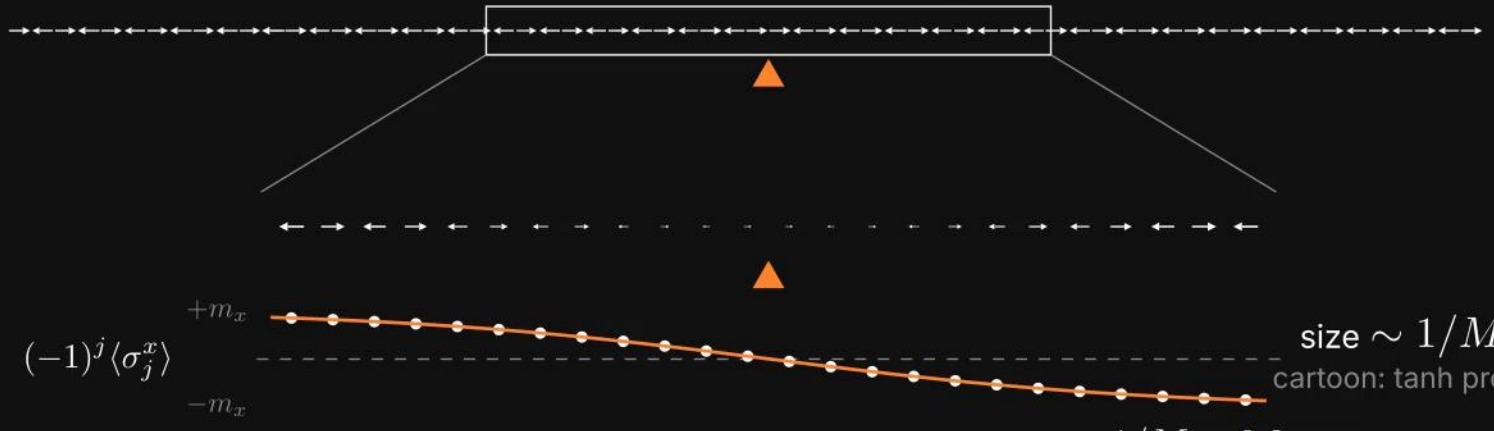
kink size $1/M$
$= 9.2$ sites

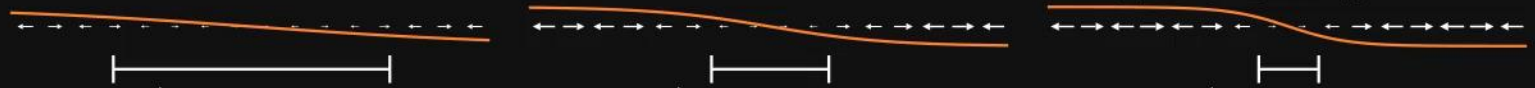
M is an inverse length:
$1/M$ counts sites

BLACK BOX · Baxter/JKM: M known exactly

- Baxter (1972) and Johnson, Krinsky and McCoy (1973) solved this chain; M follows from their formulas at any Δz, γ of the strip $|\Delta z| < 1$, $\gamma > 0$ (elliptic functions, Ledger); card again in 4.8.
- On the XY line it reduces to M = atanh γ.
- Every M in these talks is this exact value, never a fit.

a wall between the two falls of the pencil: a kink





$1/M = 9.2$ sites
$M = 0.10836$ (home)
Delta_z = 0.3, gamma = 0.2 (home)
paper: jkm_exact.py

$1/M = 3.9$ sites
$M = \operatorname{atanh} 0.25 = 0.2554$
XY line: Delta_z = 0, gamma = 0.25
closed form: M = atanh gamma

$1/M = 2$ sites
$M = 0.5$
cartoon, no coupling

M is one over a length; an energy needs a speed

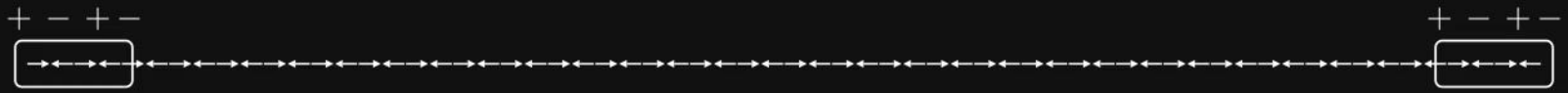
$$E_{\text{kink}} = v\,M$$

energy cost $= v\,M$, v ≈ 4 on the XY line closed form, XY line

walls: 0
closed form

▲ kink ▲ antikink

pair flight: cartoon (gen_kinks.py)
home coupling: Delta_z = 0.3, gamma = 0.2



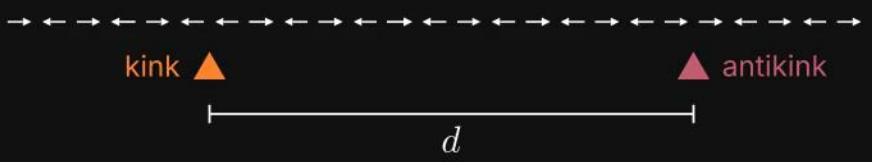
cartoon

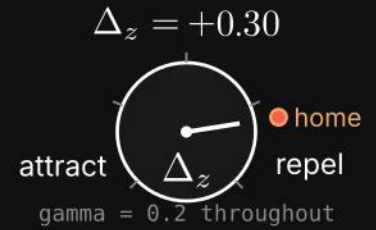
$$\sigma_j^z \left| \rightarrow\leftarrow\rightarrow\leftarrow \cdots \right\rangle \rightarrow \text{kink} + \text{antikink}$$

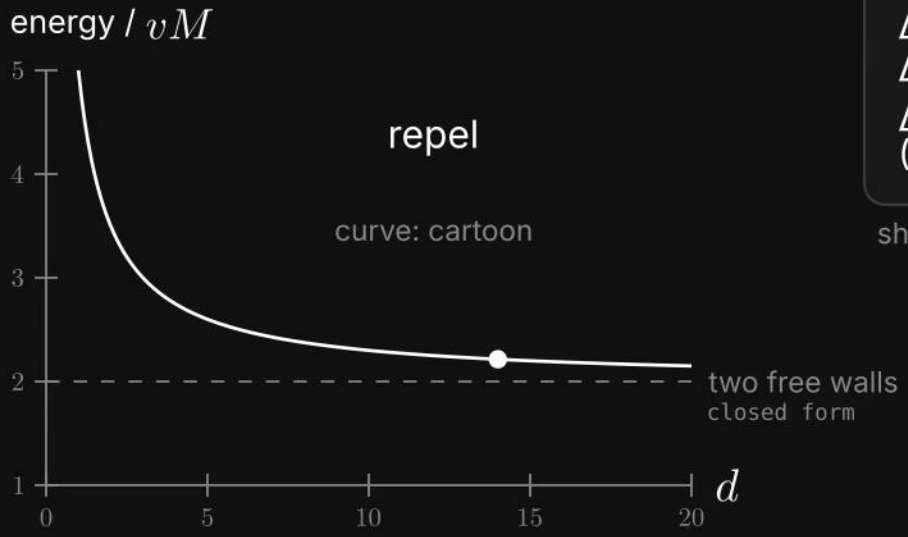
A single σ^z makes walls in pairs.

far left and far right never change; the number of walls between them is always even (Ledger)

at home, where walls repel, a pair costs $2\,vM$ closed form

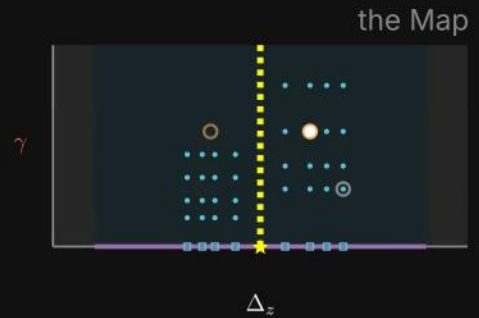






Δz < 0: attract.
Δz > 0: repel.
Δz = 0: ignore each other (free fermions).

shapes are cartoons; the sign is the physics



at ξ = 1 the pair is no longer bound

$\Delta_z = +0.00$



$\xi = 1.00$

cartoon: a free kink and antikink drift apart



$m_{B_1} = 2M \sin \dfrac{\pi\xi}{2} \quad (\xi < 1)$

two free walls and above

the bound pair $2\sin(\pi\xi/2) = 2.00$

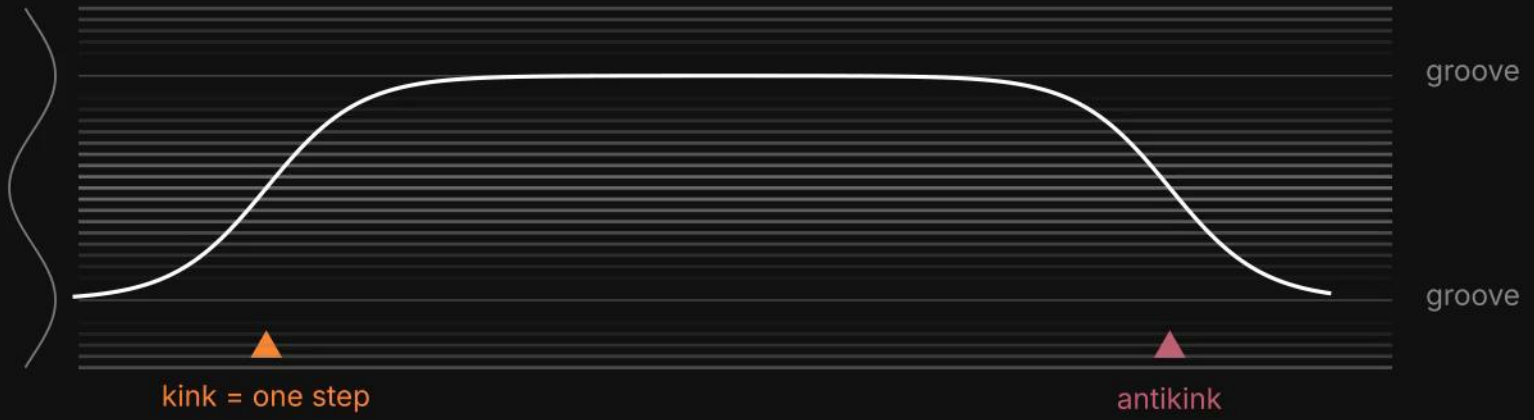
Physicists relabel Δz as ξ:
ξ = 1 free, ξ < 1 attractive, ξ > 1 repulsive.
Home: Δz = 0.3 is ξ = 1.48.

scaling-limit law; on the lattice at $\Delta_z = -0.2737,\ \gamma = 0.05$ the ratio $m_{B_1}/2M = 0.89047$ against $\sin(0.35\pi) = 0.89101$
(paper: jkm_exact.py masses_attractive)
only the first bound state exists for ξ ≥ 0.55; more appear only below ξ = 1/2



the recipe Δz → ξ is on the Ledger

paper: jkm_exact.py masses_attractive; masses.npz ratio_mB1_2M, sin_pi_xi_2, xi
Delta_z = -0.2737, gamma = 0.05; rung 2 sin(pi xi/2), knob Delta_z(xi): closed form



cartoon: the needles' angle as a string; each groove is one fall of the pencil

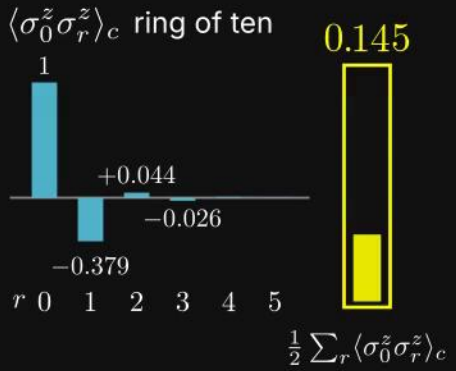
$$m_{B_1} = 2M \sin \frac{\pi\xi}{2} \quad (\xi < 1)$$

BLACK BOX · `sine-Gordon`

- As γ → 0 the stripes' angle becomes a smooth field in a cosine washboard; kinks are its steps between grooves; the recipe Δz → ξ is on the Ledger.
- The bound-pair law holds in that limit; on the lattice at Δz = −0.2737, γ = 0.05 the ratio m_B1/2M is 0.89047 against 0.89101.
- Part 4 borrows from it the bound-pair law above and what Lukyanov found for how a kink pair and the bound pair couple to the charge (4.7).

`paper: jkm_exact.py masses_attractive masses.npz ratio_mB1_2M, sin_pi_xi_2`

χzz is called a susceptibility.
For a conserved charge at temperature T, the answer to a small field is the width per site divided by T.
Here the charge is not conserved: χzz is strictly the width per site, at T = 0.



$$\chi_{zz} = 2 \lim_{\ell \to \infty} \mathrm{Var}(Q_A)/\ell = \frac{1}{2} \sum_r \langle \sigma_0^z \sigma_r^z \rangle_c$$

the variance is the bars added up, times ℓ/2, up to the edges

```
ring toy, pinned, L = 10;
Delta_z = 0.3, gamma = 0.2
(accumulator 1/2 - 0.379
+ 0.044 - 0.026 + ... = 0.145;
never joined to the iDMRG
value)
```

a ring of ten cannot hold a 4.6-site memory: a toy

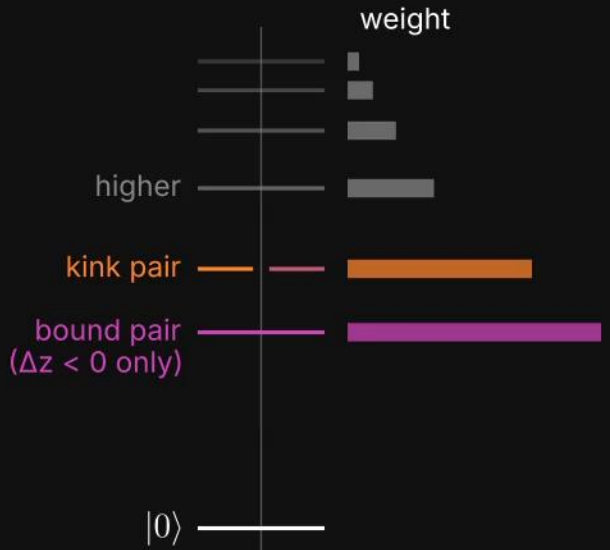
```
1/(2M) = 4.6 sites,
M = 0.10836:
paper: jkm_exact.py
```

0.1106

infinite chain (iDMRG, next scene)

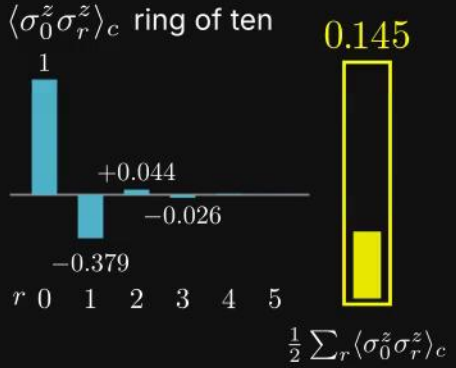
```
paper:
c_extrap_data.json
["0.3"]["0.2"][0]
= 0.110552, shown
as 0.1106; iDMRG,
infinite chain,
Delta_z = 0.3,
gamma = 0.2
```

$$\mathrm{Var}(Q_A) = \sum_{k \neq 0} |\langle k|Q_A|0\rangle|^2$$



cartoon: rungs and weights not computed;
the bound pair exists only at Delta_z < 0 (3.10b)

a sum over excited states;
Part 4 shows why the lightest weigh most



ring toy, pinned, L = 10;
Delta_z = 0.3, gamma = 0.2
(accumulator 1/2 - 0.379
+ 0.044 - 0.026 + ... = 0.145;
never joined to the iDMRG
value)

a ring of ten cannot hold
a 4.6-site memory: a toy

1/(2M) = 4.6 sites,
M = 0.10836:
paper: jkm_exact.py

0.1106

infinite chain
(iDMRG, next scene)

paper:
c_extrap_data.json
["0.3"]["0.2"][0]
= 0.110552, shown
as 0.1106; iDMRG,
infinite chain,
Delta_z = 0.3,
gamma = 0.2

closed form

2^L numbers: $2^{10} = 1{,}024$ $2^{20} = 1{,}048{,}576$ $2^{64} \approx 1.8 \times 10^{19}$



JPEG for wavefunctions.

D	χ_{zz}
64	0.033726
96	0.033723
128	0.033723

locked

paper: chi_conv.json rows[].chizz at D = 64, 96, 128
Delta_z = 0.5, gamma = 0.1, the hardest point
iDMRG, infinite chain

BLACK BOX · iDMRG (everything inside)

- The train of tables is a matrix-product state (formula on the Ledger); "grow" and "compress" are its two moves, run with TeNPy in the paper.
- Keeping the largest D eigenvalue bars at every cut is why it works for gapped chains (area law, 1.9).
- Convergence is judged by raising D until the digits stop moving; χzz is read from correlator sums in the converged state.

TAKE-HOME

$$\chi_{zz} = 2\lim \mathrm{Var}(Q_A)/\ell = \tfrac{1}{2}\sum_r \langle \sigma_0^z \sigma_r^z \rangle_c$$

- γ tilts the table (explicit, [H, Q] ≠ 0) and manufactures the red corners; the stripes are one fall of the pencil (spontaneous); the meter cares about the tilt.
- A gap buys a short memory: correlations fade over 1/(2M) sites, errors are kinks of size 1/M costing vM, made in pairs; M is exact (Baxter) and is an inverse length.
- χzz is the width of the charge bell per site: twice the slope of the variance against ℓ, or half the sum of the z correlations over every distance; 0.1106 at home from iDMRG.

0.1106: iDMRG, infinite chain; Delta_z = 0.3, gamma = 0.2

LEDGER

- ☐ Jordan-Wigner: “XY line = free fermions” taken on faith (3.1b).
- ☐ The finite ring returns the cat (both falls at once); the pinned state is the combination of the two lowest ring states with the largest overlap with the Neel-x product state (scene 3.3, scene 3.5).
- ☐ Ring energies are in units of the couplings and are never compared with M; the ladder energies do not depend on the pinning (3.4).
- ☐ Gap ⇒ exponential decay (Hastings-Koma 2006) and Baxter/JKM's exact M (elliptic functions) taken on faith; on the attractive side the fading length is one over the bound pair's mass, not 1/(2M) (3.7).
- ☐ A kink's energy is vM; v ≈ 4 on the XY line from the free-fermion dispersion; the value at Δz = 0.3 is not pinned yet (3.8).

v ≈ 4: closed form, XY line (free-fermion dispersion)

LEDGER

- ☐ “A single σ^z makes walls in pairs”: the signed count of walls is conserved because the stripes at infinity cannot change (topological charge) (3.9).
- ☐ Recipe (formula below, left) gives ξ from Δz; home $\xi = 1.4813$; the bound-pair law is the $\gamma \to 0$ limit; only B_1 exists for $\xi \geq 0.55$ (3.10b).
- ☐ $\chi zz = 0.110552$ at home (c_extrap_data.json); the accumulator 0.145 (ring toy, pinned, L = 10) is a toy and never joins it; the 3.11 word said once, as width per site at T = 0; for this non-conserved charge the answer to a small field is a different number at every temperature (3.11).
- ☐ “Variance per site” is the talk's name for χzz, a deliberate simplification: χzz is twice the long-window limit of the charge variance divided by ℓ, so the variance per site is strictly $\chi zz/2 = 0.0553$ (3.11).
- ☐ Matrix-product state (formula below, right): one D × D matrix per site, the amplitude is their trace; convergence judged in D at $\Delta z = 0.5$, $\gamma = 0.1$ (3.12).

$$\theta_0 = \arccos \Delta_z, \quad \beta^2 = 1 - \theta_0/\pi, \quad \xi = \beta^2/(1-\beta^2) \qquad |\Psi\rangle = \sum_s \mathrm{Tr}[A^{s_1} A^{s_2} \cdots]\, |s_1 s_2 \cdots\rangle$$

0.110552: paper: c_extrap_data.json; iDMRG, infinite chain
0.0553 = 0.110552/2: closed form
0.145: ring toy, pinned, L = 10; Delta_z = 0.3, gamma = 0.2
xi = 1.4813: closed form (the recipe) at Delta_z = 0.3

$$\Delta S_A^{(2)} = \tfrac{1}{2}\log(4\pi V), \quad V \stackrel{?}{=} \tfrac{1}{2}\chi_{zz}\ell$$

χ_{zz} (measured): 0.1106

M (inverse size, exact): 0.10836

The chain has a measurable variance per site χzz and an exactly known inverse size M. Is V = χzz ℓ/2 for a mixed piece? Can the lightest particles bound χzz/M from below with nothing fitted?

chi_zz: iDMRG, infinite chain
M: paper: jkm_exact.py
Delta_z = 0.3, gamma = 0.2